

Abstract

Egypt is currently facing various challenges in effectively utilizing its resources, particularly in the areas of recycling and alternative energy sources. In response to these challenges, the present study proposes a novel approach to using Egypt's water wealth, from rivers and lakes to the two important seas that Egypt overlooks. To achieve that purpose, the capstone challenge is improving alternative energy usage. In addition to other challenges, which are slightly dependent on this challenge, such as climate change and improving the usage of arid areas. So, the ideal, solution for this problem is harnessing the kinetic energy of water for generating electricity. The prototype's mechanism starts with the basic principle behind electromagnetic induction. So, the prototype was constructed as a system that consists of a floating part and an immersed part, which consists of three units; each unit is made from magnets in a small tube with a coil around it. When the floating part moves with the waves, the magnets move too and produce a magnetic field, which makes electricity. By implementing this approach, the project aims to decrease the challenges facing Egypt in terms of sustainable resource management and promote a more eco-friendly and responsible approach toward energy consumption. For results, the potential difference was between 0.9 to 1.4 volts, and the power (watt) was also calculated to ensure the effectiveness of the prototype and how it achieved the design requirements.

Introduction

Like many other developing nations, Egypt faces significant problems and challenges. One of them is the energy problem and how to provide alternative sources. The country has a high demand for energy due to its growing population, yet it struggles to meet this demand with its limited domestic energy resources. This has resulted in a heavy reliance on imported fossil fuels, which not only puts a strain on the economy but also harms the environment, which causes another problem, leading to the next challenge: reducing the effect of climate change. For this reason, alternative energy sources have become a topic of interest in Egypt. Alternative sources of energy are defined as any source of energy that is not derived from fossil fuels, such as solar, wind, and geothermal energy. These sources of energy are often renewable and have lower carbon emissions compared to traditional sources of energy, making them an attractive alternative to address the Egyptian energy crisis. One example of an alternative is water energy, which refers to energy harnessed from the flow and movement of water in the form of hydropower, tidal power, and wave power. So, our solution focuses on providing alternative energy sources by generating electricity from the kinetic energy of the water, especially wave motion. The electricity that is produced can be used in arid areas. For example, the new agricultural projects supplement conventional sources of electrical grids and help solve another challenge, which is to improve the use of arid areas. During the research, a method for harnessing the wave motion, which are point absorbers, was found. Point absorbers are a type of wave energy converter technology that converts the kinetic energy of ocean waves into electricity. It works by absorbing energy from ocean waves through a buoy or floating part that moves up and down with the waves and is connected to a generator to produce electricity(De Rijcke, 2011). One of their advantages is that they have a low environmental impact, as they do not require dams or large reservoirs that can harm fish and other aquatic species. On the other hand, one of its disadvantages is its high cost. Point absorbers are an expensive technology due to the high cost of materials used in their construction. Because of this point, action must be taken to modify this solution. So, the proposed solution depends on using recycling and widespread materials. The solution's main idea of working is electromagnetic induction which has the ability to convert kinetic energy directly into electrical energy. our Materials are permanent magnets that will move with the waves in and out of a that will be wrapped with a copper coil, which will cut the magnetic field which and produce pipe electricity. Finally, the design requirements of the prototype were effectiveness and to produce of a satisfactory amount of electricity. The prototype has been made to be effective, as every part performs its desired function, and the results have been as expected.

Materials & Methods

Material	Amount	Usage	Description	Price	Source	Picture
1 Neodymi un	36 pieces	Magnetic fields can be used to make electricity	long, circular magnets where the length exceeds the diameter Length=0.32 Diameter=1.23	5 L.E	Recycled from old speakers and from a shop for magnets	
2 Floating body	One	Used to help the magnet to float and to move with the waves	Its density of 0.1 is lower than the density of water	50 L.E	Swimming g tools store	
3 Copper wire	0.75kg	is wrapped around the three tubes with a total number of turns of 3000 coils	Insulated copper wire with diameter= 3mm for 1 kg	400 L.E	Electronic store	
4 Capacitor	Three pieces	Used as a device for storing electrical energy	Electronic device with two arms (positive, negative)	5 L.E	Electronic store	
5 Bridge rectifier	Three pieces	converts an oscillating current into a single-directional direct current (DC)	Electronic device with AC converter to DC and two arms (positive and negative)	5 L.E	Electronic store	

Total: 993 L.E

Methods

The prototype consists of two main parts, a fixed part, and a moving part.

The fixed part:

- A plastic PVC tube was cut using a saw into three pieces, every piece with dimensions of 20cm in height and 4.5 cm in diameter.
- A copper wire was wrapped 3000 turns around each pipe using the motor winding machine
- The three pipes were connected to each other in a serious connection after connecting each pipe with a capacitor and rectifier which all were connected together with a PCB board using a welding machine.

To make insulant:

- An acrylic box was made with the dimensions of 25 cm in height, 25 cm in width, and 25 cm in length. It was cut by a laser cutter in the form of a puzzle Pieces to make sure that no water entered.

The three pipes and the PCB were fixed in the box.

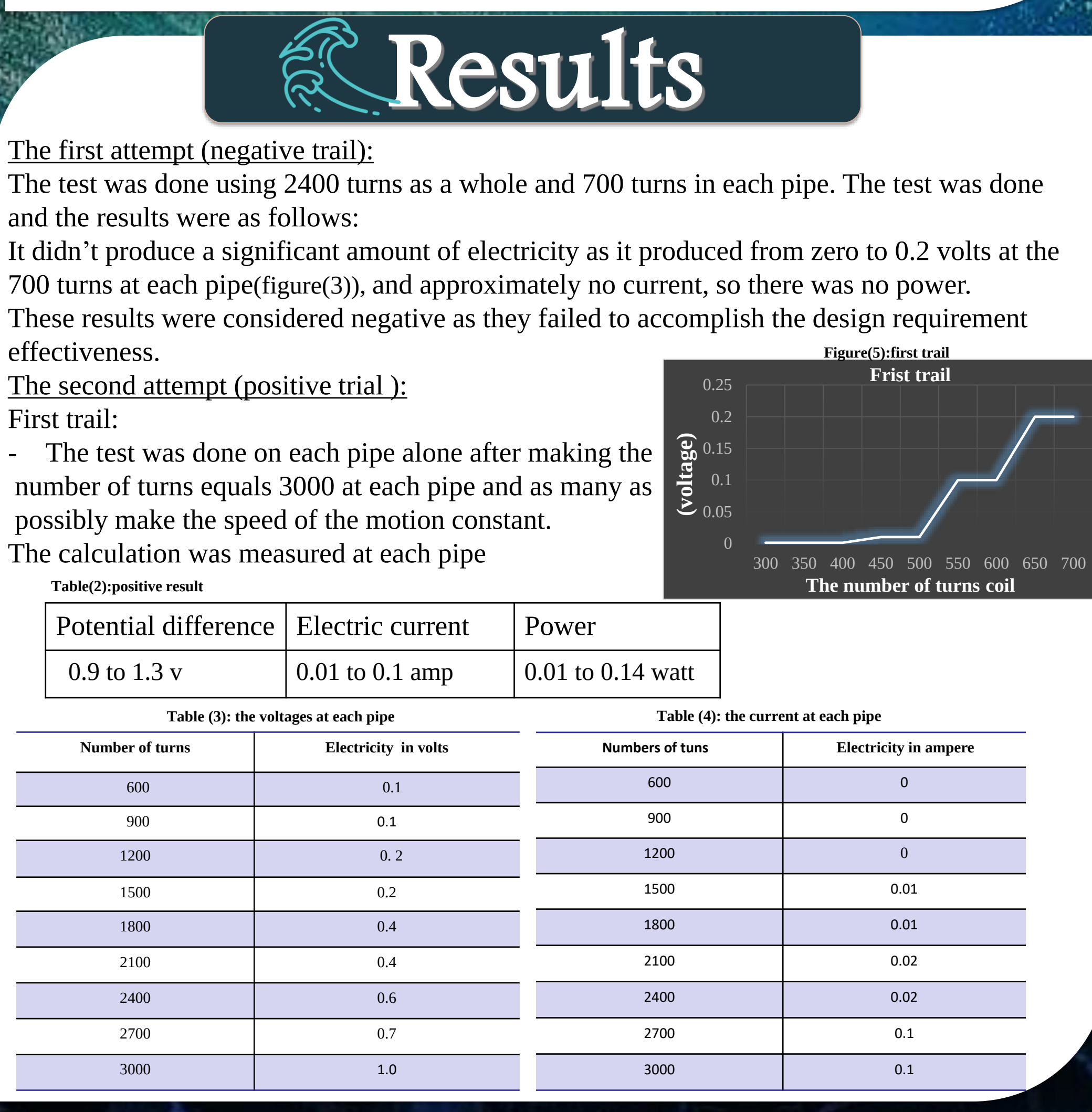
The moving part:

- Three magnets, in each pipe were attached to a string that is connected to a lightweight that floats on water (the floating body) , which was attached to the other end of the string, so whenever the waves move the floating body, the magnet can move up and down freely inside the pipe so it can generate electricity, this step was done in every pipe.
- The upper side of the acrylic box was pierced with three holes using the laser cutter, and the string connecting the weight and the magnet passes through each hole. The diameter of each hole was 7 mm.

A plastic container with dimensions of length =50cm,height=30cm, and width =35cm was filled with water to represent the seawater in the largescale.

The test plan was conducted to accomplish the design requirements which are:

- Effectiveness: This requirement was tested by making sure that each part of the prototype works at first, such as making sure that the magnet is doing its desirable function which is generating electricity from the kinetic energy of water. This will be calculated by the amount of electricity. It will be tested by a multimeter to calculate the potential difference (V), and the electric current (I) to calculate the power in watts that equal to V*I. A light bulb is also used to indicate the amount of electricity that is produced, as the light bulb needs 1.5 volts to illuminate and 15 milliamperes. That is to make sure that the prototype will generate a suitable amount of electricity.



Analysis

Egypt is one of the largest Arab countries with a rapidly growing population, which makes the energy consumption rate high. The country has been facing a serious energy crisis regarding how to provide energy resources for years now. Energy resources are divided into renewable resources and nonrenewable resources, as it is known in (ES.1.09). (Benbow, Carpenter, Hoover, Smith, & Southard, 2012) A renewable resource is a resource that can be replaced in nature at a rate close to its rate of use, while a nonrenewable resource is a resource that exists in a fixed amount or is used faster than it can be replaced in nature. These renewable energy sources are referred to as "alternatives" because they represent an alternative to coal, oil, and natural gas, on which Egypt is highly dependent, especially natural gas and oil.

On the other hand, Egypt possesses an abundance of land, sunny weather, and high wind speeds, making it a prime location for renewable energy projects. So, our solution solves this problem by producing renewable and clean energy by harnessing the wave energy of seawater to generate electricity. Waves are in constant motion and therefore possess a consistent form of energy. In our model wave-energy generator, the idea of working is dependent on electromagnetic induction. Electromagnetic induction is the production of an electric current in a conductor when there is a change in the magnetic field around it. This phenomenon was discovered by Michael Faraday in the 19th century (National High Magnetic Field Laboratory's Magnet Academy, 2022). The basic principle behind electromagnetic induction is that a magnetic field produces an electric current. When a conductor is placed in a magnetic field, if the magnetic field changes, then an electric current is induced in the conductor. This is because the changing magnetic field induces a voltage in the conductor, which causes the electrons in the conductor to move.

producing an electric current. According to all of this, our solution is mainly on changing the magnetic field to produce electricity.

We used recycled materials; the main materials are the neodymium magnet, copper wire, and plastic PVC pipes. This neodymium magnet is considered a rare-earth magnet. Rare-earth magnets are permanent magnets composed of the alloy's neodymium, iron, and boron. The main advantage of it is that it does not require a continuous electric supply, it only loses 1% of its power in 10 years.

These magnetic energy devices provide pollution-free energy and will not deplete our natural resources. The prototype consists of 3 systems each system has about 12 magnets from rare earth magnets. The force of attraction of each magnet is 52N. Copper wire is mainly used as a conductor because of its excellent electrical conductivity, which is one of the most essential properties required for the efficient flow of electricity. It can conduct electricity more effectively compared to other materials like aluminum, steel, or gold. It's high conductivity and good tensile strength also mean that it can withstand wear and tear better than other materials, making it a more durable choice for application in generators, as it was studied in (CH. 1.09) (Zumdahl & Zumdahl, 2014).In addition to all this, a capacitor and a rectifier were used to convert AC current into DC current and store electric current. The goal of their use is to achieve design requirements that will achieve satisfactory effectiveness and energy.

Several tests were carried out to reach the desired goals.

-The first test was done with the number of turns of copper wire around each pipe equal to 700 turns to make the total number of turns equal to 2400 turns, and the number of magnets that were used in each system was six magnets. The numbers of turns were very low, and the power of the magnetic field was too weak to produce a significant amount of electricity. After some research, we found that the induced electric current can be increased by increasing the strength of the magnetic field, increasing the speed of the change in the magnetic field, or increasing the number of turns of the copper coil.

-The second test was done after some modifications, and the number of turns of the copper coil around each pipe increased to 3000 turns each, so the total became 90000 turns in the whole system.

The result was that each pipe produced a voltage range between 0.9 to 1.3 volts and current range between 0.01 to 0.1 ampere that also changed according to the wave velocity.

One of the most important calculations in generating electricity is Calculating the power of electricity. The power of electricity is calculated in determine the rate at which electrical energy is consumed or generated. Power is measured in watts and is calculated by multiplying the voltage (in volts) by the current (in amperes). The power of the final prototype equals v* amps = watts. According to the law the power of each pipe approximately 0.14 watts.

These results achieved the design requirements:

- Effectiveness: As a lot of modifications were done on the prototype ,the number of the coil that helps make the prototype effective in achieving the desired goals .As our prototype is in the water, we must consider what Archimedes' principle states as it studied in (PH.1.08) : an object immersed in a fluid experiences a buoyant force that is equal in magnitude to the force of gravity on the displaced fluid. The buoyant force is an upward force exerted on objects submerged in fluids that equals o the weight of the fluid that the body displaced. $F(\text{buoyant}) = \rho g V$, =the density of water, which is 1000 kg/m3, g = the gravity (9.8 m/s2), and V = the immersed part. (Serway & Jewett, 2013) .The Mediterranean Sea is a potential location for a floating buoy in Egypt due to its high wave energy levels, average wave heights of 1-2 meters, and wave frequencies of 7-10 seconds.in addition, it is generally colder than the Red Sea, which can affect electromagnetic induction. High temperatures can cause the resistance of a wire or coil to increase, which can decrease the efficiency of an electromagnetic induction process.

Conclusions

After the prototype was constructed, testing was done, and data was collected, some conclusions were made. According to the test analysis, when increasing the number of turns of copper coil around each pipe, the amount of electricity will increase, by making the power of the magnetic field a controlling variable as each system has 12 magnets. The amount of electricity was tested after making 3000 turns of coil around each pipe. According to the prior solution of using wave motion to get electricity, a point absorber was found, which requires a high cost of materials used in their construction. Therefore, there must be a solution to this issue. in our solution, recycling materials were used as recycling PVC pipes to be the body where the coil will be wrapped. Based on the data above, it's obvious that the design requirements have been met, as the prototype would be effective that each part of the prototype has done its desirable function which is to generate electricity by the kinetic energy of water and produce voltage = 0.9 to 1.3, ampere = 0.01 to 0.1 and power = 0.01 to 0.14 in each pipe

Recommendation

It is recommended for any future study that will work on this project to:

- The system can be covered with solar cells that can convert the light energy of the sun into electricity.
- In the future and on a large scale, it is recommended to use a stronger magnet.
- The material of the acrylic box is better changed into a stronger material like titanium to be more suitable for the sea environment.
- Another potential recommendation for increasing the electricity generated by the buoy system could be to incorporate a wind turbine. Wind turbines can generate electricity from the kinetic energy of the wind and could be mounted on the buoy system to take advantage of the wind energy available in the area.

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